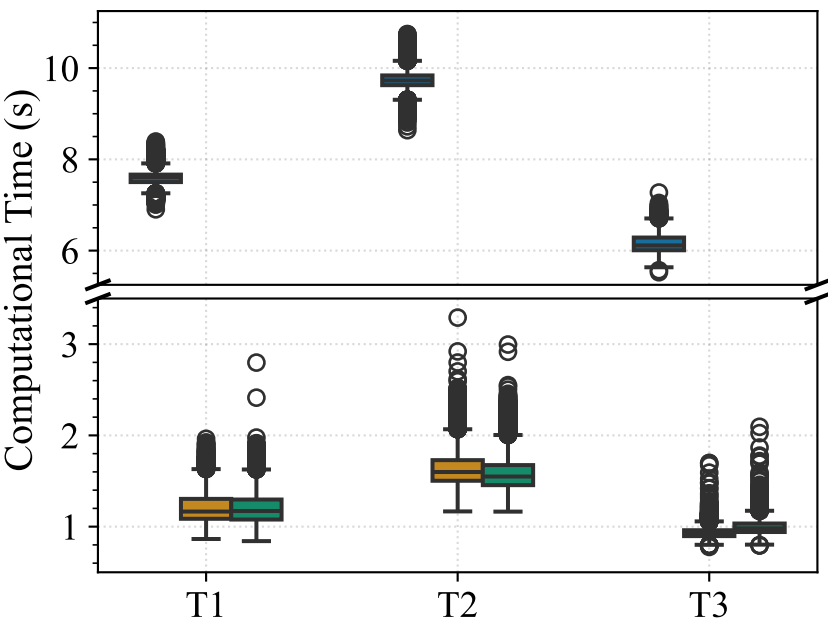
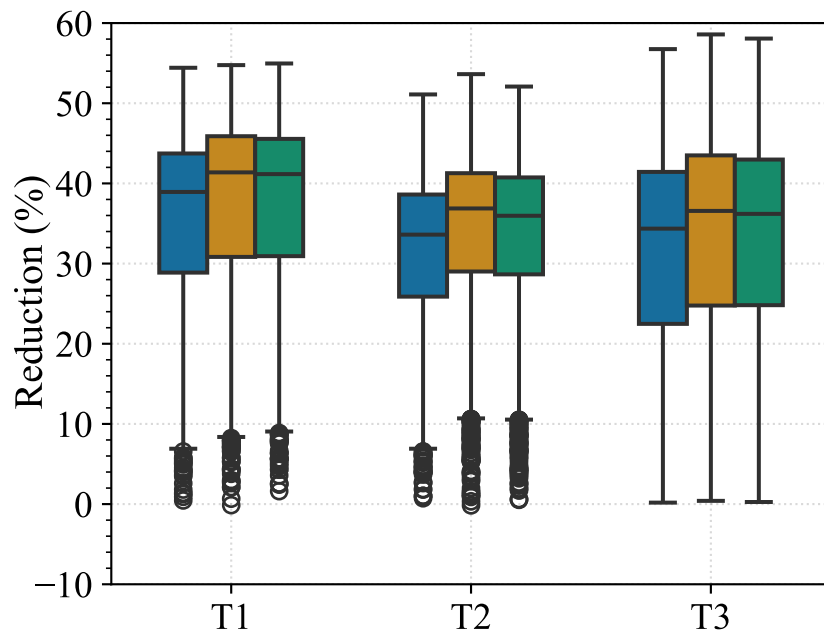
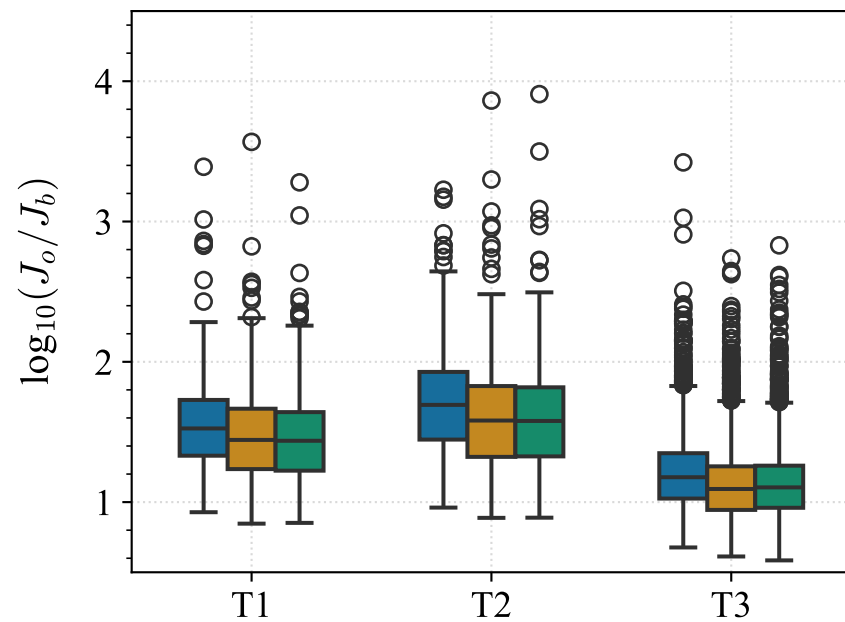


(a) 4D-Var Computational Time**(b) Cost Function Reduction****(c) Observation-Background Ratio**

Twin Experiment



Traditional-NPZ



PI-NPZ



NN-NPZ